

CSE273:FOUNDATIONS OF MACHINE LEARNING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: Explain the fundamentals of Machine Learning, its types, and differences from AI and Data Science.

CO2 :: Apply Python libraries (Pandas, Matplotlib) to load, inspect, manipulate, and visualize datasets for meaningful insights.

CO3 :: Apply exploratory data analysis techniques (univariate and bivariate plots, correlation analysis) to identify basic patterns and trends in datasets.

CO4 :: Apply linear algebra, probability, and calculus concepts to interpret the mathematical foundations of machine learning algorithms.

CO5 :: Apply statistical techniques and optimization methods to analyze data and understand model training and convergence behavior in machine learning.

CO6 :: Understand how machine learning models learn, generalize, and are chosen based on data

Unit I

Machine Learning Foundation : Introduction to Machine Learning, Types of Machine Learning, Difference between ML, AI and Data Science, Parametric vs Non-parametric models, Generative vs Discriminative Models, Machine Learning Workflow, Train-Test data split, Overview of Evaluation Metrics, Overfitting and Underfitting, Bias-Variance Trade-off, Applications and Use-cases of ML

Numpy for Numerical Computing : Creating Arrays, Operations on Arrays, Broadcasting Rule, Matrix operations, Random Module

Unit II

Pandas for Data Handling : Series and Data Frame, Reading and writing CSV file, Inspecting Datasets, Selecting and filtering data

Data Visualization using Matplotlib : Line plot, Bar plot, Histogram, Scatter plot, Pie chart, Customizing plots, Subplots

Unit III

Exploratory Data Analysis : Types of data, Loading data using Pandas, Univariate analysis using Histogram, Box plot, Count plots, Bivariate analysis using Scatter plots, Line plots, Violin plots for category vs numeric, Correlation analysis using Heatmaps, Multicollinearity, Distribution analysis: Skewness, Kurtosis, Detecting patterns, anomalies and trends, EDA workflow integration

Unit IV

Linear Algebra and Calculus : Scalars, Vectors, Matrices, Tensors, Eigenvalues and Eigenvectors

Probability Foundations : Random variables, Probability distribution, Mean, Variance, Covariance, Joint, Marginal and Conditional Probability, Baye's Theorem, Likelihood vs Probability

Calculus for ML : Functions, Gradient, Partial derivatives, Chain Rule

Unit V

Statistics for ML : Sampling, Population, Hypothesis Testing, Confidence Intervals, Correlation, Maximum Likelihood Estimation

Optimization Methods : Loss Functions, Convex vs Non-convex Functions, Learning Rate Impact, Local Minima, Global Minima, Saddle Points, Gradient Descent, Stochastic Gradient Descent (SGD), Advanced optimizers (Momentum, RMSProp, Adam)

Unit VI

Statistical Learning Theory : Well-Posed Learning Problems, Components of a Learning System, Statistical Learning Framework, Empirical Risk Minimization, Inductive Bias, Role of Prior Knowledge, Probably Approximately Correct Learning, Sample Complexity, No Free Lunch Theorem, Choosing Algorithms Based on Data and Assumptions

List of Practicals / Experiments:

List of Practicals

- 01. Write a program using NumPy to create and manipulate arrays, perform vectorized operations, apply broadcasting, and execute basic matrix computations.

- 02. Load a CSV dataset and explore its structure using Pandas, and create appropriate visualizations to understand data distribution and relationships using Matplotlib.
- 03. Perform Exploratory Data Analysis on the Titanic dataset to study passenger demographics, survival patterns, distribution of numerical features, and relationships between categorical and numerical variables.
- 04. Conduct EDA on the Students Performance dataset to analyze score distributions, the impact of parental education and study hours on student performance, and correlations among numerical attributes.
- 05. Implement vector and matrix operations using NumPy, compute eigenvalues and eigenvectors, and numerically calculate the gradient of a given multivariable function using partial derivatives.
- 06. Simulate random variables from different probability distributions and compute mean, variance, covariance, and conditional probability.
- 07. Using the Students Performance dataset, draw samples to estimate population parameters, compute correlations, construct confidence intervals, and perform a hypothesis test on mean score differences.
- 08. Using the Boston Housing dataset, implement and compare Gradient Descent and its variants (SGD, Momentum, RMSProp, Adam) on the regression loss function to study learning rate impact and convergence behavior.
- 09. Write a program using NumPy to create and manipulate arrays, perform vectorized operations, apply broadcasting, and execute basic matrix computations.
- 10. Write a NumPy program to generate a random 3×3 matrix and perform transpose, trace, rank, matrix multiplication with its transpose, and inverse computation (if possible)

Text Books:

1. MACHINE LEARNING USING PYTHON by MANARANJAN PRADHAN, U DINESH KUMAR, WILEY

References:

1. MACHINE LEARNING IN DATA SCIENCE USING PYTHON by DR. R. NAGESWARA RAO, WILEY
2. MACHINE LEARNING by ANURADHA SRINIVASARAGHAVAN, VINCY JOSEPH, WILEY
3. DATA SCIENCE AND MACHINE LEARNING USING PYTHON by REEMA THAREJA, MC GRAW HILL
4. REAL-WORLD MACHINE LEARNING by HENRIK BRINK, JOSEPH W. RICHARDS, MARK-FETHEROLF, BEAU CRONIN, WILEY
5. MATHEMATICS FOR MACHINE LEARNING by MARC PETER DEISENROTH, A. ALDO FAISAL, CHENG SOON ONG, CAMBRIDGE UNIVERSITY PRESS